Figure 2.2 The Natter database schema consists of social spaces and messages within those spaces. Spaces have an owner and a name, while messages have an author, the text of the message, and the time at which the message was sent. Unique IDs for messages and spaces are generated automatically using SQL sequences.

Using your favorite editor or IDE, create a file schema.sql under natter-api/src/main/resources and copy the contents of listing 2.2 into it. It includes a table named `spaces` for keeping track of social spaces and their owners. A sequence is used to allocate unique IDs for spaces. If you've not used a sequence before, it is a bit like a special table that returns a new value every time you read from it.

Another table, `messages`, keeps track of individual messages sent to a space, along with who the author was, when it was sent, and so on. We index this table by time, so that you can quickly search for new messages that have been posted to a space since a user last logged on.

Listing 2.2 The database schema – schema.sql

```
CREATE TABLE spaces( #A
space_id INT PRIMARY KEY,
name VARCHAR(255) NOT NULL,
owner VARCHAR(30) NOT NULL
);
CREATE SEQUENCE space_id_seq; #B
CREATE TABLE messages( #C
space_id INT NOT NULL REFERENCES spaces(space_id),
msg_id INT PRIMARY KEY,
author VARCHAR(30) NOT NULL,
```